



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°4_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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NIVEAU: L3

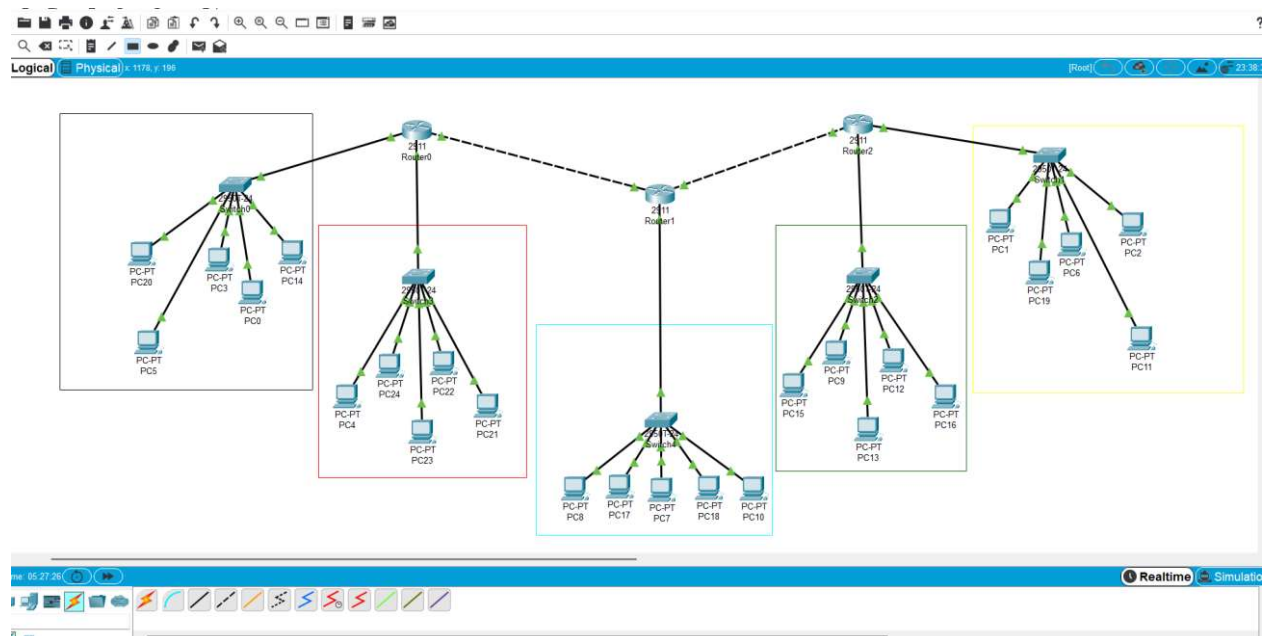
DATE: 19/04/2026

L'objectif de ce TD est de:

- 1-Comprendre le rôle des protocoles de routage dynamique
- 2-Configurer RIP v2, OSPF, EIGRP
- 3-Comparer leur comportement
- 4-Tester la convergence et la connectivité
- 5-Simuler des pannes réseau

- 1- Concevoir un réseau de campus universitaire
- 2- Mettre en place une topologie multi-routeurs
- 3- Organiser les utilisateurs par services
- 4- Comprendre et configurer un protocole de routage dynamique
- 5- Tester la connectivité entre les différents réseaux

Création de la topologie



Configuration IP de base

R0

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router0
Router0(config)#
Router0(config)#interface g0/0
Router0(config-if)#ip address 192.168.10.1 255.255.255.0
Router0(config-if)#no shutdown

Router0(config-if)#
Router0(config-if)#interface g0/1
Router0(config-if)#ip address 192.168.20.1 255.255.255.0
Router0(config-if)#no shutdown

Router0(config-if)#
Router0(config-if)#interface g0/2
Router0(config-if)#ip address 10.0.0.1 255.255.255.252
Router0(config-if)#no shutdown

Router0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
```

R2

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#
Router1(config)#interface g0/0
Router1(config-if)#ip address 192.168.30.1 255.255.255.0
Router1(config-if)#no shutdown

Router1(config-if)#
Router1(config-if)#interface g0/1
Router1(config-if)#ip address 10.0.0.2 255.255.255.252
Router1(config-if)#no shutdown

Router1(config-if)#
Router1(config-if)#interface g0/2
Router1(config-if)#ip address 10.0.0.5 255.255.255.252
Router1(config-if)#no shutdown

Router1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

```

Ctrl+F6 to exit CLI focus

R2

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router2
Router2(config)#
Router2(config)#interface g0/0
Router2(config-if)#ip address 192.168.40.1 255.255.255.0
Router2(config-if)#no shutdown

Router2(config-if)#
Router2(config-if)#interface g0/1
Router2(config-if)#ip address 192.168.50.1 255.255.255.0
Router2(config-if)#no shutdown

Router2(config-if)#
Router2(config-if)#interface g0/2
Router2(config-if)#ip address 10.0.0.6 255.255.255.252
Router2(config-if)#no shutdown

Router2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

```

Ctrl+F6 to exit CLI focus

Configuration RIP v2

R1

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#
Router0(config)#router rip
Router0(config-router)#version 2
Router0(config-router)#no auto-summary
Router0(config-router)#network 192.168.10.0
Router0(config-router)#network 192.168.20.0
Router0(config-router)#network 10.0.0.0
Router0(config-router)#
```

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R2

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#
Router1(config)#router rip
Router1(config-router)#version 2
Router1(config-router)#no auto-summary
Router1(config-router)#network 192.168.30.0
Router1(config-router)#network 10.0.0.0
Router1(config-router)#network 10.0.0.4
Router1(config-router)#
```

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R3

```
Router2>enable
Router2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#
Router2(config)#router rip
Router2(config-router)#version 2
Router2(config-router)#no auto-summary
Router2(config-router)#network 192.168.40.0
Router2(config-router)#network 192.168.50.0
Router2(config-router)#network 10.0.0.4
Router2(config-router)#
```

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Configuration DHCP sur les routeurs pour attribuer des address ip automatique

```
Router0>en
Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#ip dhcp pool ADMIN
Router0(dhcp-config)#network 192.168.10.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.10.1
Router0(dhcp-config)#dns-server 8.8.8.8
Router0(dhcp-config)#
Router0(dhcp-config)#ip dhcp pool ETUDIANTS
Router0(dhcp-config)#network 192.168.20.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.20.1
Router0(dhcp-config)#dns-server 8.8.8.8
Router0(dhcp-config)#
```

Ctrl+F6 to exit CLI focus

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☐ Top

PC7

PC18

PC10

```
Router1>en
Router1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#ip dhcp pool PROFESSEURS
Router1(dhcp-config)#network 192.168.30.0 255.255.255.0
Router1(dhcp-config)#default-router 192.168.30.1
Router1(dhcp-config)#dns-server 8.8.8.8
Router1(dhcp-config)#
```

Ctrl+F6 to exit CLI focus

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PC8

PC17

PC7

PC18

PC10

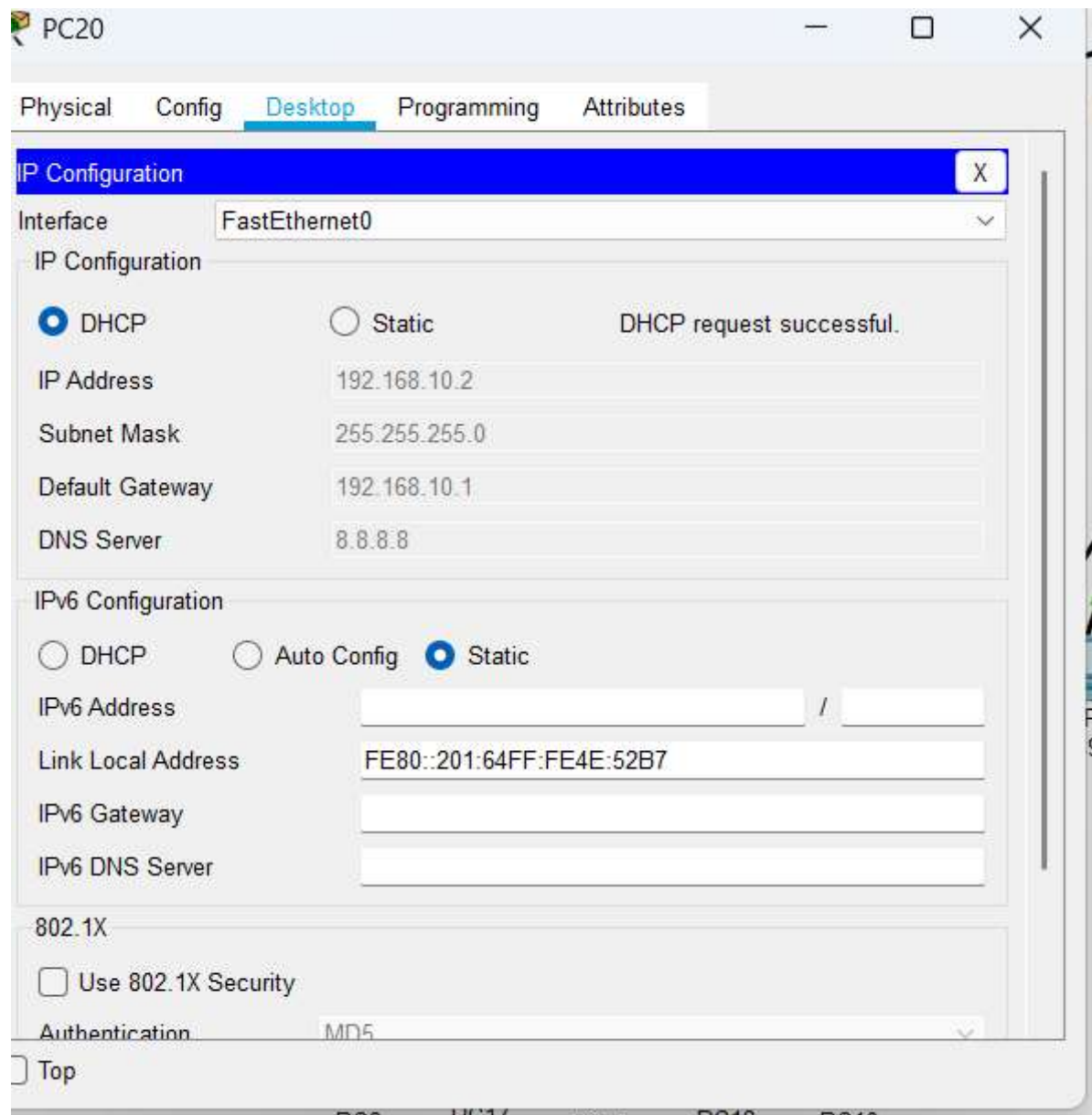
```
Router2>en
Router2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#ip dhcp pool BIBLIO
Router2(dhcp-config)#network 192.168.40.0 255.255.255.0
Router2(dhcp-config)#default-router 192.168.40.1
Router2(dhcp-config)#dns-server 8.8.8.8
Router2(dhcp-config)#
Router2(dhcp-config)#ip dhcp pool VISITEURS
Router2(dhcp-config)#network 192.168.50.0 255.255.255.0
Router2(dhcp-config)#default-router 192.168.50.1
Router2(dhcp-config)#dns-server 8.8.8.8
Router2(dhcp-config)#
```

Ctrl+F6 to exit CLI focus

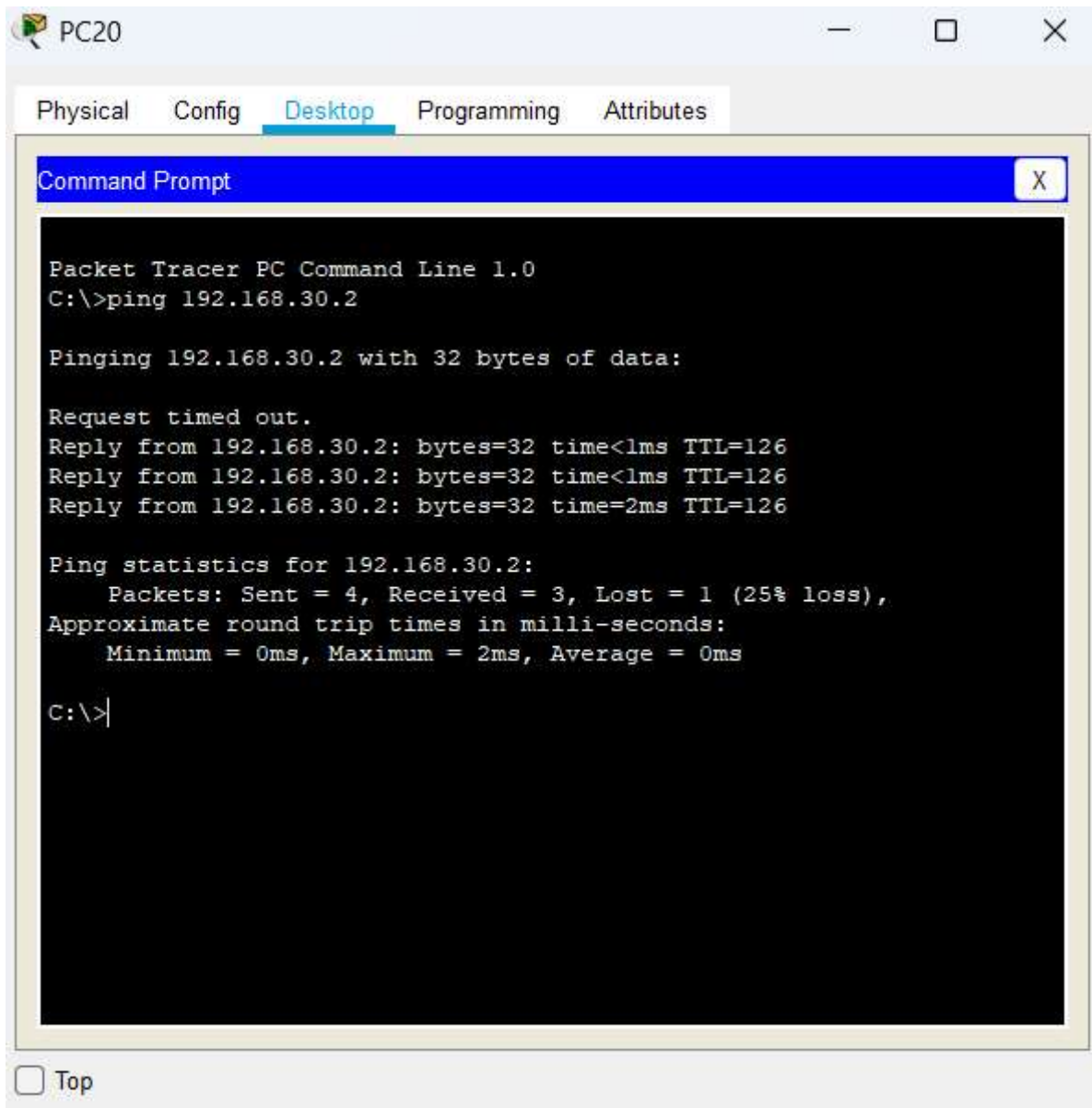
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Exemple d'un PC avec address ip configuré avec DHCP



Tests RIP avec ping sur des PC de VLAN different



```

Router0>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:02, GigabitEthernet0/2
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
  192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
R       192.168.30.0/24 [120/1] via 10.0.0.2, 00:00:02, GigabitEthernet0/2
R       192.168.40.0/24 [120/2] via 10.0.0.2, 00:00:02, GigabitEthernet0/2
--More--

```

Ctrl+F6 to exit CLI focus

Coupure d'un lien

```

Router0>en
Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#interface g0/1
Router0(config-if)#shutdown

```

Ctrl+F6 to exit CLI focus

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```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Après on etablie la connection a nouveau

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Router0>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:26,
GigabitEthernet0/2
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.30.0/24 [120/1] via 10.0.0.2, 00:00:26,

```

Ctrl+F6 to exit CLI focus

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☐ Top

Configuration OSPF

R1

Router0(config-if)#

Router0(config-if)#router ospf 1

Router0(config-router)#network 192.168.10.0 0.0.0.255 area 0

Router0(config-router)#network 192.168.20.0 0.0.0.255 area 0

Router0(config-router)#network 10.0.0.0 0.0.0.3 area 0

Router0(config-router)#

Ctrl+F6 to exit CLI focus

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☐ Top

R2

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#
Router1(config)#router ospf 1
Router1(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router1(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router1(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router1(config-router)#
```

Ctrl+F6 to exit CLI focus

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R3

```
Router2>enable
Router2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#
Router2(config)#router ospf 1
Router2(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router2(config-router)#network 192.168.50.0 0.0.0.255 area 0
Router2(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router2(config-router)#
```

Ctrl+F6 to exit CLI focus

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Tests OSPF

```
Router1>show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address
Interface				
192.168.20.1	1	FULL/DR	00:00:39	10.0.0.1
GigabitEthernet0/1				
192.168.50.1	1	FULL/DR	00:00:39	10.0.0.6
GigabitEthernet0/2				

```
Router1>
```

Ctrl+F6 to exit CLI focus

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Coupure d'un lien

Enter configuration commands, one per line. End with CNTL/Z.

```
Router1(config)#interface g0/2
```

```
Router1(config-if)#shutdown
```

```
Router1(config-if)#
```

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to  
administratively down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface  
GigabitEthernet0/2, changed state to down
```

```
04:22:15: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.50.1 on  
GigabitEthernet0/2 from FULL to DOWN, Neighbor Down: Interface  
down or detached
```

Ctrl+F6 to exit CLI focus

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Vérification

Physical Config CLI Attributes

IOS Command Line Interface

```
Router2#
%SYS-5-CONFIG_I: Configured from console by console

Router2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0
    192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/1
L       192.168.50.1/32 is directly connected, GigabitEthernet0/1
```

Ctrl+F6 to exit CLI focus

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Top

Configuration EIGRP

R0

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#
Router0(config)#router eigrp 100
Router0(config-router)#network 192.168.10.0
Router0(config-router)#network 192.168.20.0
Router0(config-router)#network 10.0.0.0
Router0(config-router)#no auto-summary
Router0(config-router)#
```

Ctrl+F6 to exit CLI focus

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R1

```
Router1#enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#
Router1(config)#router eigrp 100
Router1(config-router)#network 192.168.30.0
Router1(config-router)#network 10.0.0.0
Router1(config-router)#network 10.0.0.4
Router1(config-router)#no auto-summary
Router1(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.1
(GigabitEthernet0/1) is up: new adjacency
```

Ctrl+F6 to exit CLI focus

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R2

```

Router2>enable
Router2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router2(config)#
Router2(config)#router eigrp 100
Router2(config-router)#network 192.168.40.0
Router2(config-router)#network 192.168.50.0
Router2(config-router)#network 10.0.0.4
Router2(config-router)#no auto-summary
Router2(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.5
(GigabitEthernet0/2) is up: new adjacency

```

Ctrl+F6 to exit CLI focus

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Tests EIGRP

```

Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT   RTO  Q  Seq
                               (sec)          (ms)                  Cnt  Num
0   10.0.0.2           Gig0/2        11   00:02:42    40    1000  0   9

Router0#

```

Ctrl+F6 to exit CLI focus

```
Router1#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
```

H	Address	Interface	Hold	Uptime	SRTT	RTO	Q
Seq			(sec)		(ms)		Cnt
Num							
0	10.0.0.1	Gig0/1	13	00:03:50	40	1000	0
9							
1	10.0.0.6	Gig0/2	14	00:03:25	40	1000	0
9							

```
Router1#
```

Ctrl+F6 to exit CLI focus

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```
Router2#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
```

H	Address	Interface	Hold	Uptime	SRTT	RTO	Q
Seq			(sec)		(ms)		Cnt
Num							
0	10.0.0.5	Gig0/2	11	00:04:22	40	1000	0
10							

```
Router2#
```

Ctrl+F6 to exit CLI focus

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☐ Top

```

Router0#
Router0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
D       10.0.0.4/30 [90/3072] via 10.0.0.2, 00:05:25, GigabitEthernet0/2
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
D       192.168.30.0/24 [90/5376] via 10.0.0.2, 00:05:25, GigabitEthernet0/2
D       192.168.40.0/24 [90/5632] via 10.0.0.2, 00:05:00, GigabitEthernet0/2
--More--

```

Ctrl+F6 to exit CLI focus

```

C:\>ping 192.168.50.5

Pinging 192.168.50.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.50.5: bytes=32 time<1ms TTL=125
Reply from 192.168.50.5: bytes=32 time<1ms TTL=125
Reply from 192.168.50.5: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.50.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

Top

Questions

1- Pourquoi OSPF est plus efficace que RIP ?

OSPF est plus efficace que RIP parce qu'il utilise l'algorithme de Dijkstra pour calculer les routes, ce qui lui permet de converger rapidement et de gérer de grands réseaux sans la limite de 15 sauts imposée par RIP. De plus, OSPF organise le réseau en zones hiérarchiques, ce qui optimise la gestion et réduit le trafic inutile.

2- Quelle différence entre protocole classful et classless ?

Un protocole classful fonctionne avec des classes d'adresses fixes (A, B, C) et ne transmet pas le masque de sous-réseau, ce qui entraîne un gaspillage d'adresses IP et une faible flexibilité. En revanche, un protocole classless inclut le masque dans ses mises à jour et permet l'utilisation de sous-réseaux de tailles variables (VLSM), ce qui rend l'adressage beaucoup plus efficace et adapté aux réseaux modernes.

3- Pourquoi EIGRP est plus rapide ?

EIGRP est plus rapide parce qu'il utilise l'algorithme DUAL, qui permet de trouver immédiatement une route alternative en cas de problème. Il envoie uniquement des mises à jour partielles au lieu de retransmettre toute la table de routage, et sa métrique prend en compte plusieurs paramètres comme la bande passante, le délai et la fiabilité, ce qui le rend plus précis et réactif que RIP.

4- Quel protocole choisir pour une grande entreprise et pourquoi ?

Pour une grande entreprise, le protocole recommandé est OSPF, car il est un standard ouvert, compatible avec différents équipements, et capable de gérer des réseaux complexes grâce à sa structure hiérarchique en zones. Il offre une convergence rapide et fiable, ce qui le rend idéal pour des infrastructures de grande taille.

Cocclusion.

Ce TD m'a permis de comprendre le rôle des protocoles de routage dynamique, savoir comment configurer RIP v2, OSPF, EIGRP, .donc la tache a été réussie.